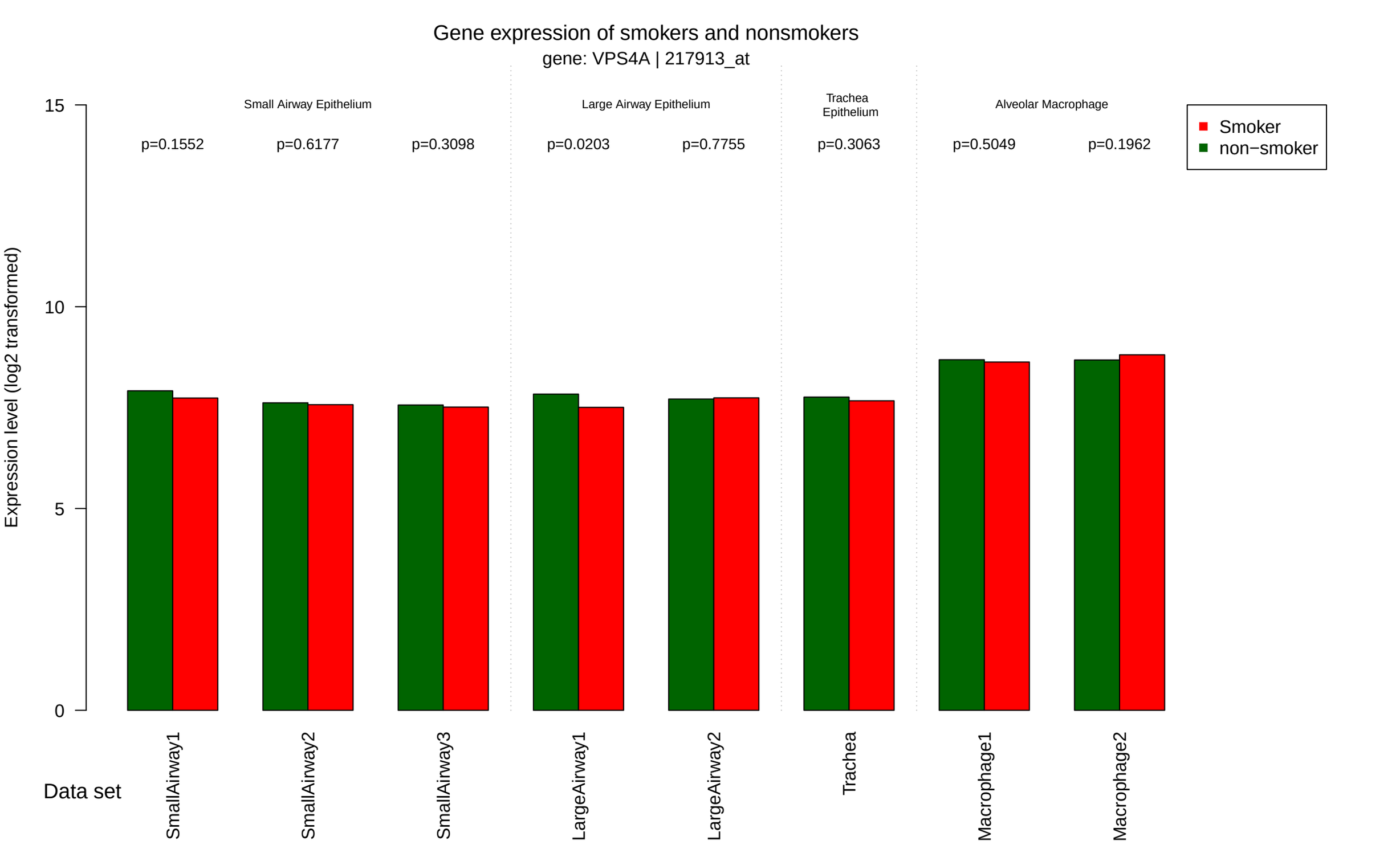
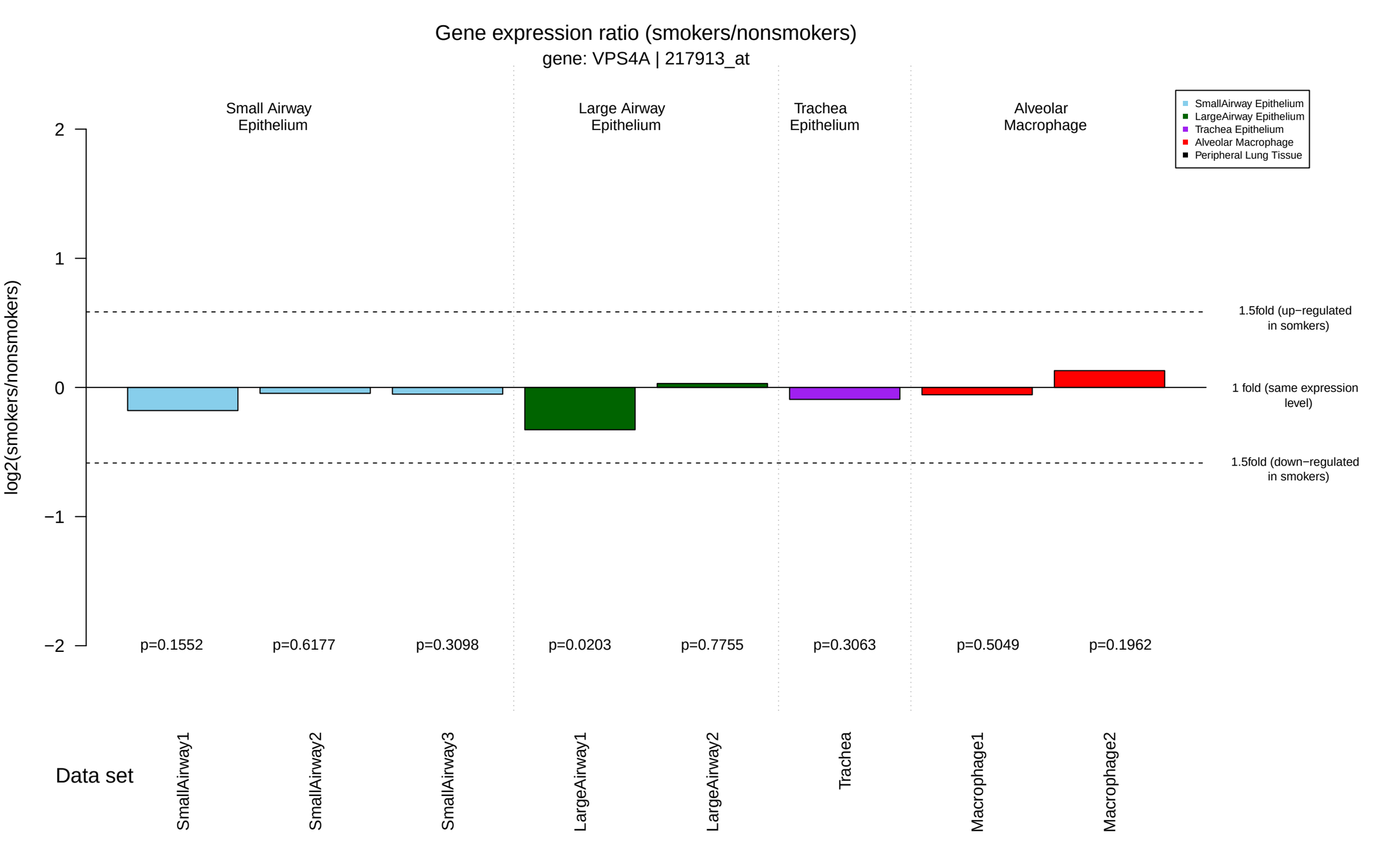
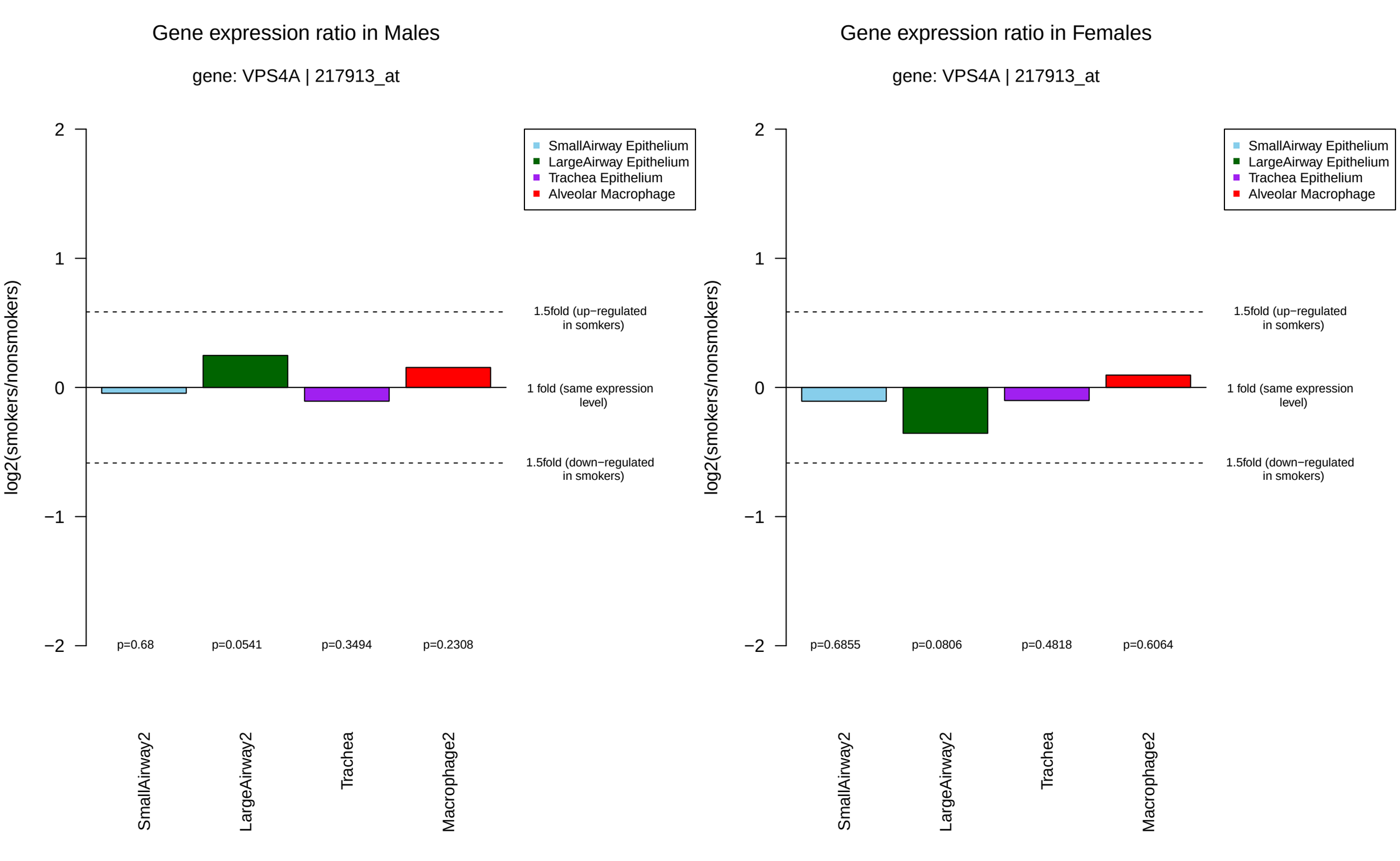
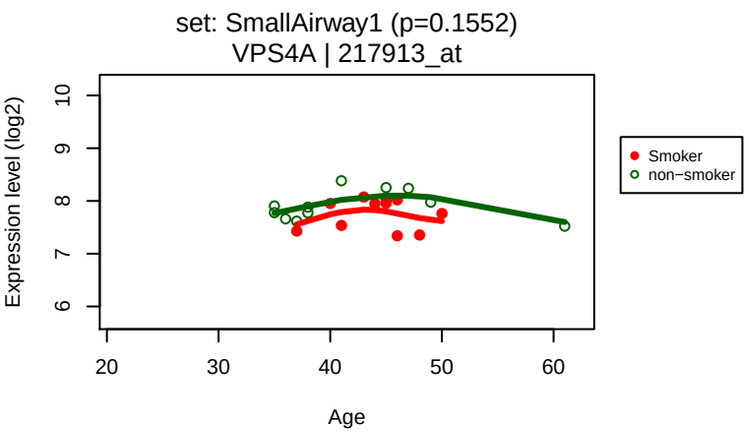
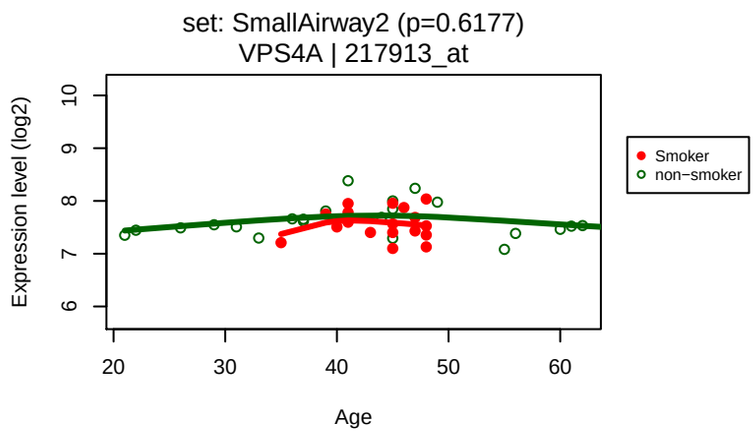
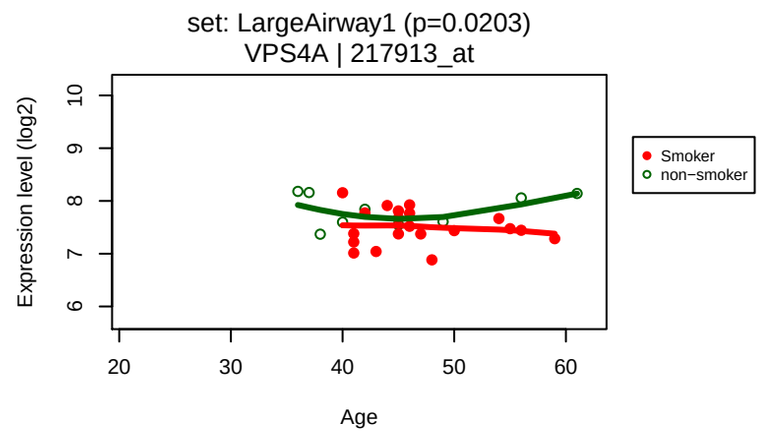

Gene expression of smokers and nonsmokers
gene: VPS4A | 217913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1552
p=0.6177
p=0.3098
p=0.0203
p=0.7755
p=0.3063
p=0.5049
p=0.1962
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

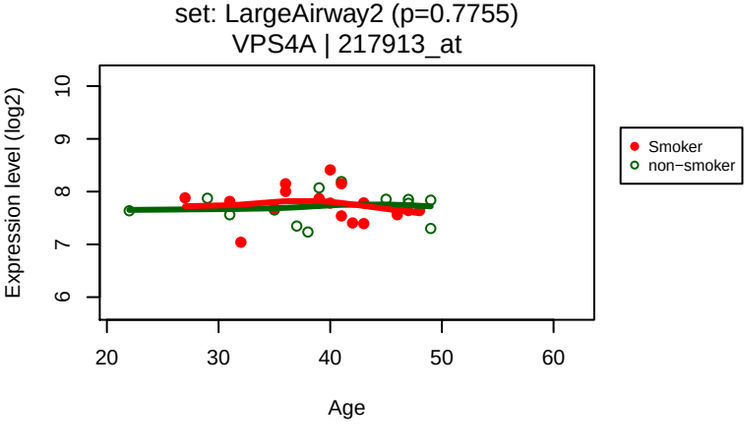

Gene expression ratio (smokers/nonsmokers)
gene: VPS4A | 217913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1552
p=0.6177
p=0.3098
p=0.0203
p=0.7755
p=0.3063
p=0.5049
p=0.1962
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

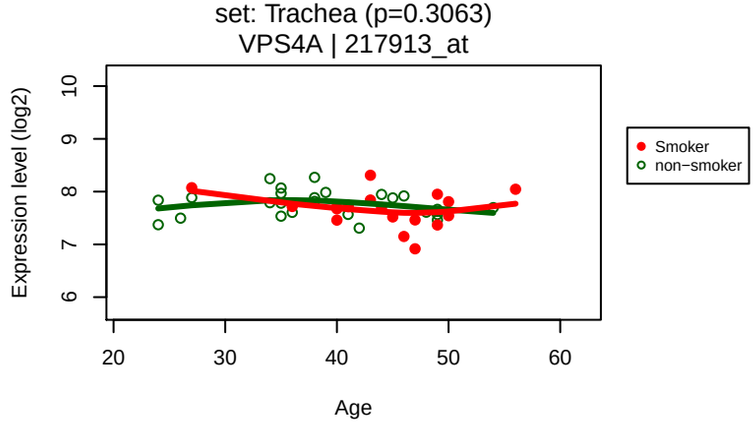

Gene expression ratio in Males
gene: VPS4A | 217913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.68
p=0.0541
p=0.3494
p=0.2308
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VPS4A | 217913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6855
p=0.0806
p=0.4818
p=0.6064
SmallAirway2
LargeAirway2
Trachea
Macrophage2

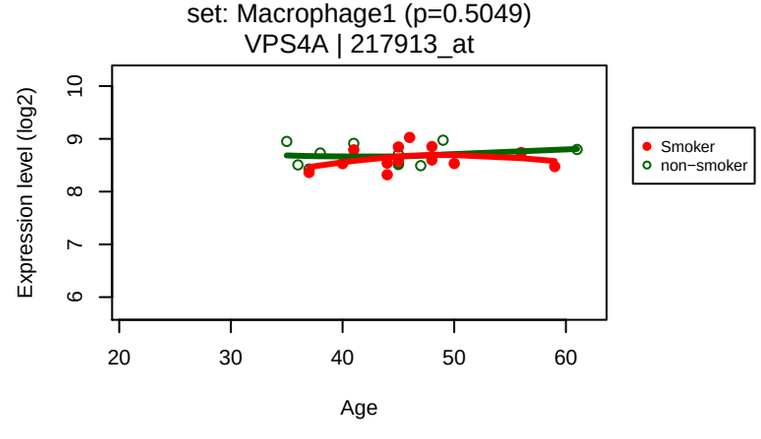

set: SmallAirway1 (p=0.1552)
VPS4A | 217913_at
Expression level (log2)
Age
Smoker
non−smoker

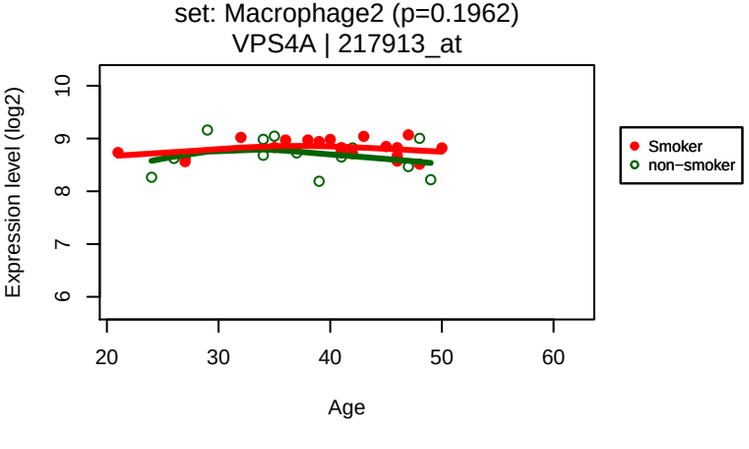

set: SmallAirway2 (p=0.6177)
VPS4A | 217913_at
Expression level (log2)
Age
Smoker
non−smoker

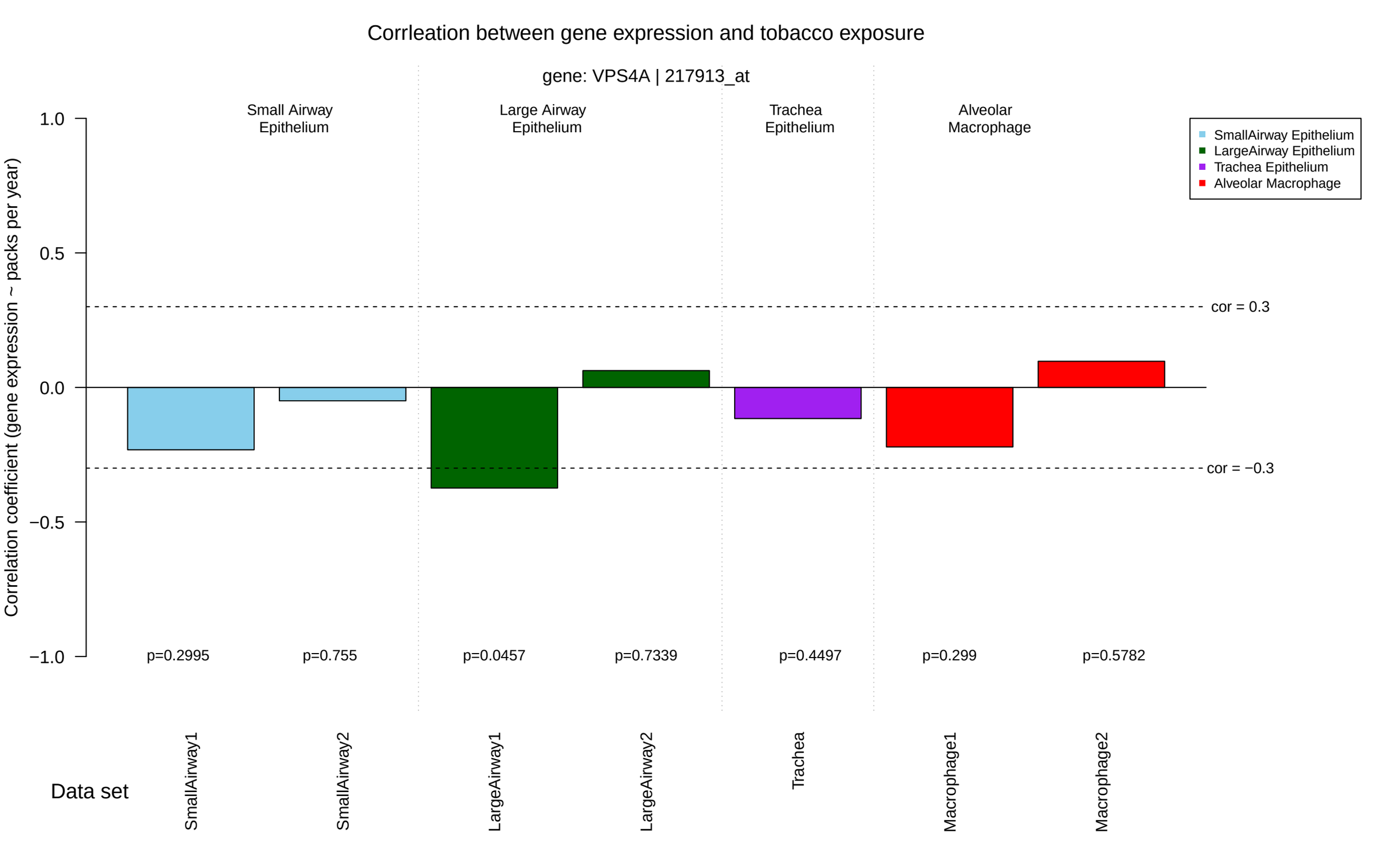

set: LargeAirway1 (p=0.0203)
VPS4A | 217913_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7755)
VPS4A | 217913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3063)
VPS4A | 217913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5049)
VPS4A | 217913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1962)
VPS4A | 217913_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VPS4A | 217913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2995
p=0.755
p=0.0457
p=0.7339
p=0.4497
p=0.299
p=0.5782
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2